Statistics for Data Science -1

Normal Distribution

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Learning objectives

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1. The pdf of Normal distribution

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2. Approximation rule

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3. Standard Normal distribution- use of Normal probability tables.

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6. Percentiles of Normal distributions.

Introduction

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- This distribution is one the most important in all statistics, both from a theoretical and application point of view.

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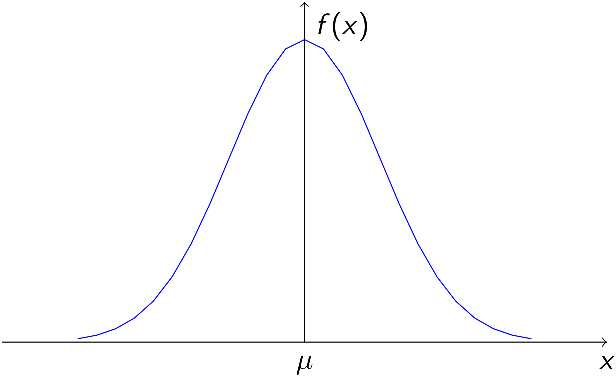

Height of curve

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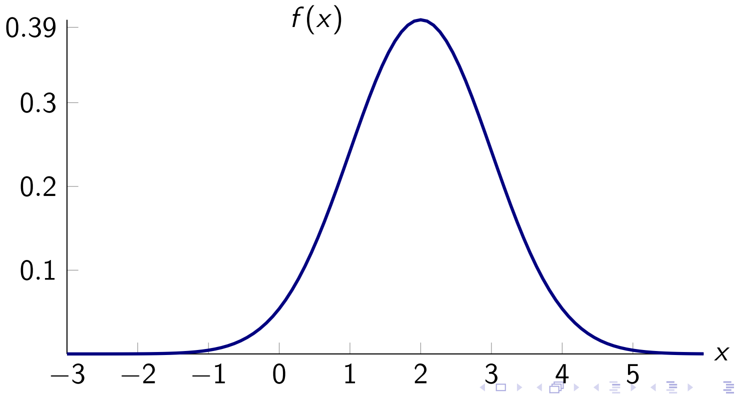

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$$\frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}; \mu = 2, \sigma = 0.5,$$

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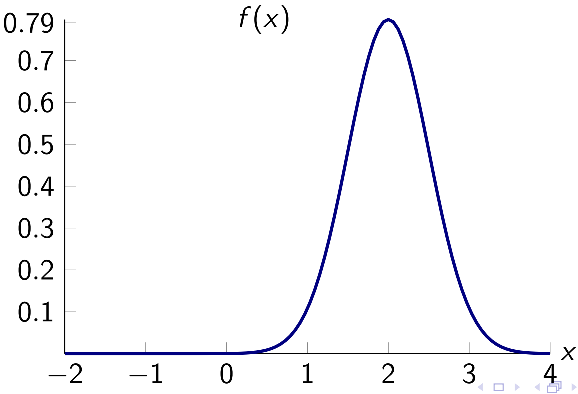
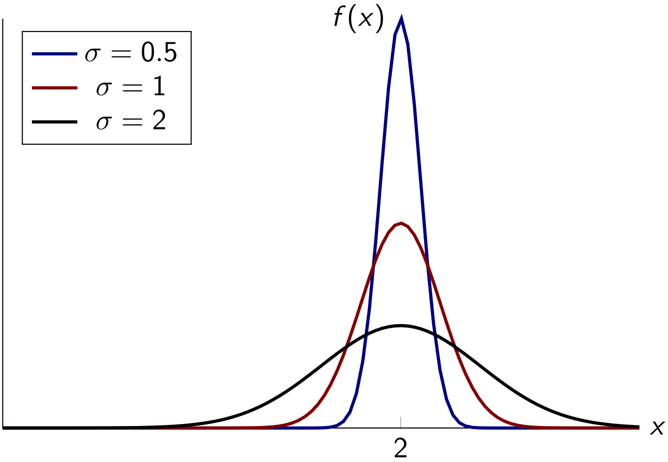
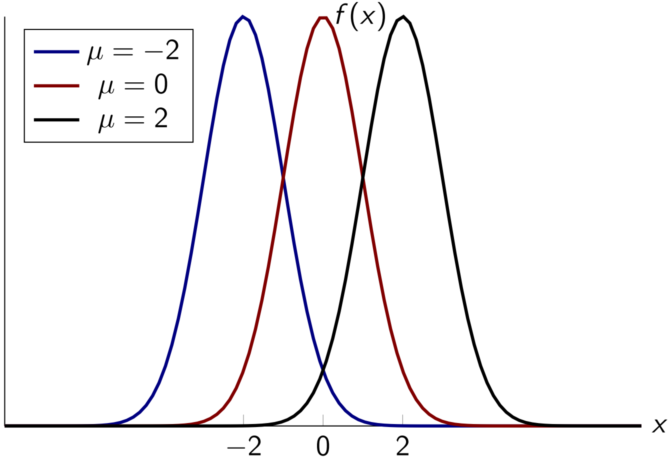


Same μ different σ



The larger the σ, the flatter and more spread out is the distribution.

Same σ different μ



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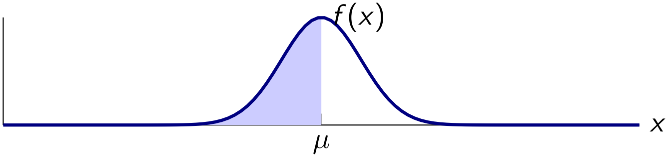

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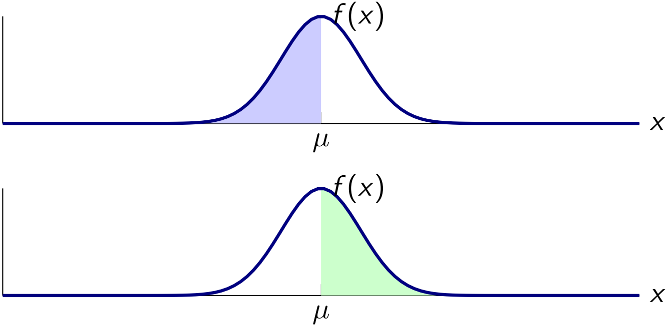

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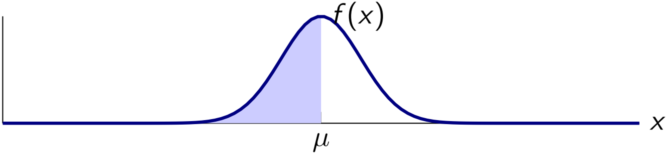

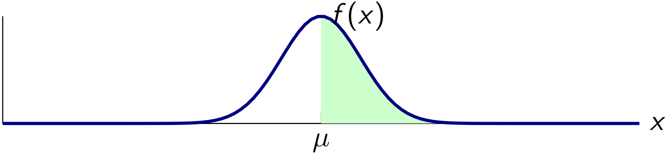

$$P(X \leq \mu) = P(X \geq \mu) = \frac{1}{2}$$

General μ and σ

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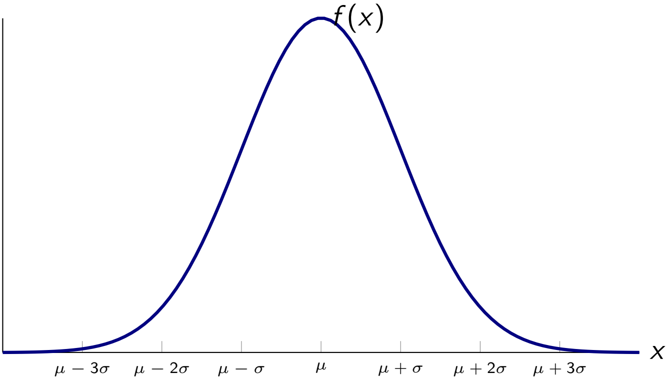

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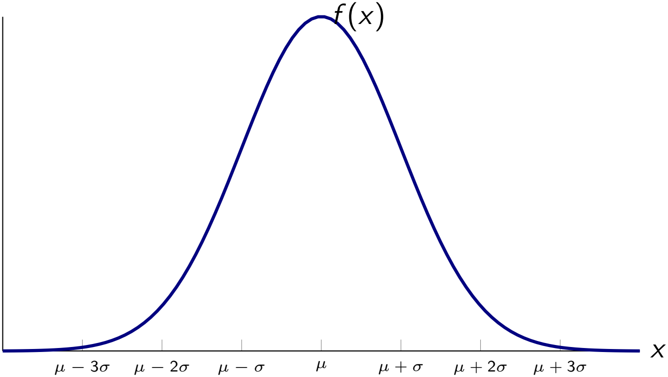

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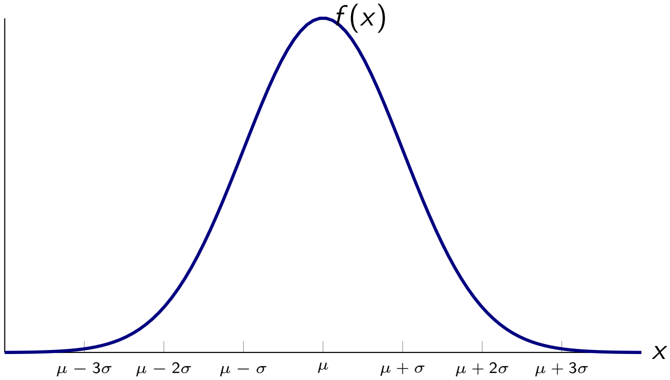
General μ and σ



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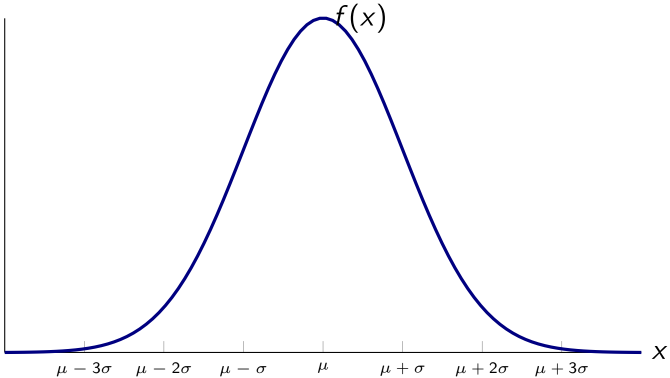


1. bell shaped,
2. centered at μ and
3. close to the horizontal axis outside the range from $\mu - 3\sigma$ to $\mu + 3\sigma$

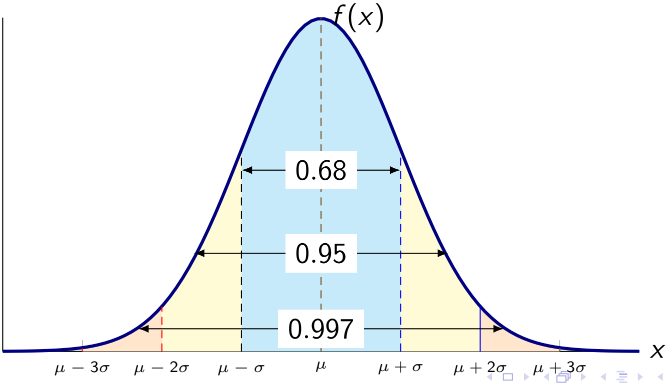
Section summary

- pdf of a Normal distribution with parameters μ and σ
- Height of the curve

Approximation rule

A normal random variable with mean μ and standard deviation σ will be

- Between $\mu - \sigma$ and $\mu + \sigma$ with approx. probability 0.68
- Between $\mu - 2\sigma$ and $\mu + 2\sigma$ with approx. probability 0.95
- Between $\mu - 3\sigma$ and $\mu + 3\sigma$ with approx. probability 0.997



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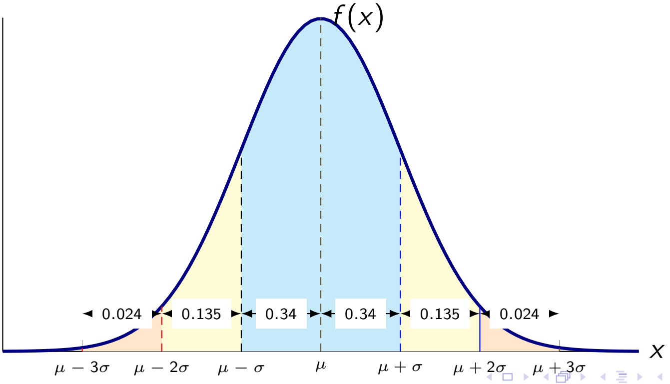

Application of approximation rule

The heights of a certain population of males are normally distributed with mean 69 inches and standard deviation 6.5 inches. Approximate the proportion of this population whose height is less than 82 inches.[1]

[1] Ross, Sheldon M. Introductory statistics. Academic Press, 2017.

Application of approximation rule

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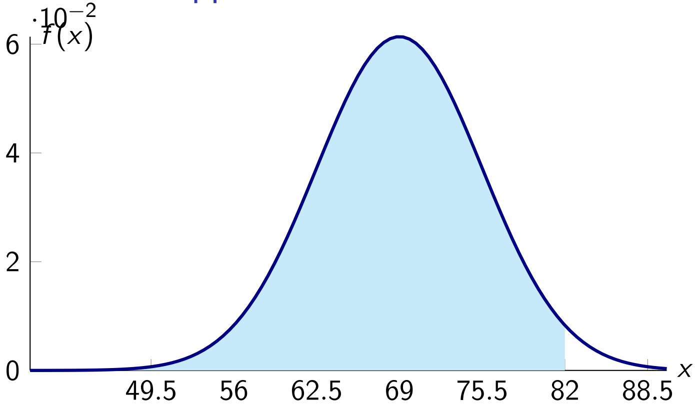
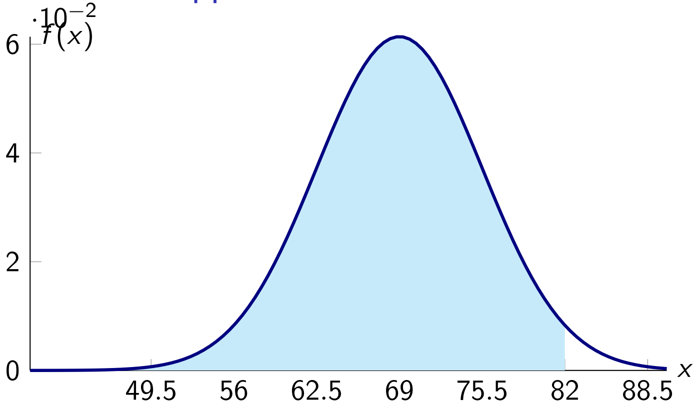
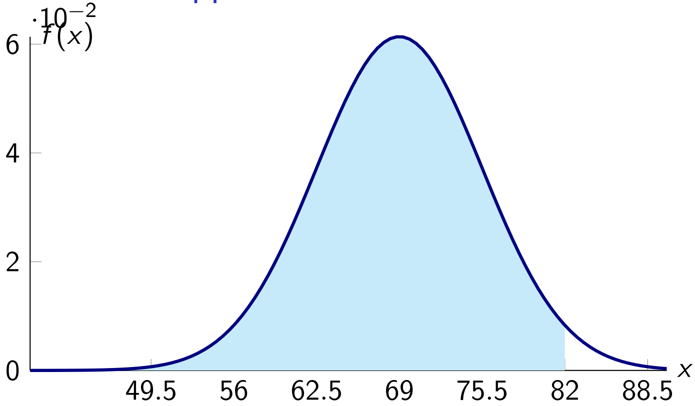


Application of approximation rule



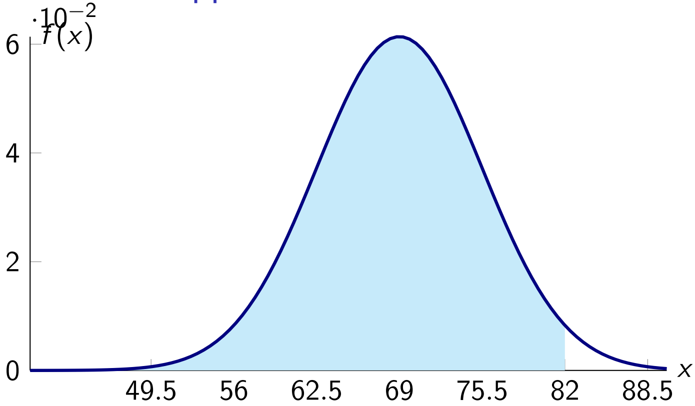
- Let X be the height of a male in inches.

Application of approximation rule



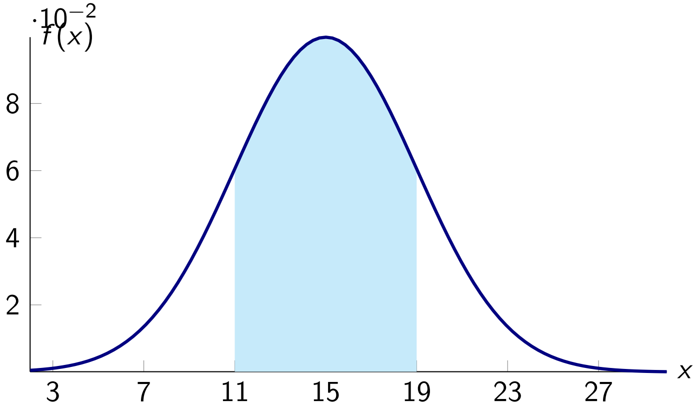
- Let X be the height of a male in inches.
- A height of 82 inches is two standard deviations above the mean. Hence area to the right of 82 inches is approx. 0.025

Application of approximation rule



- Let X be the height of a male in inches.
- A height of 82 inches is two standard deviations above the mean. Hence area to the right of 82 inches is approx. 0.025 Hence, $P(X < 82) = 1 - 0.024 = 0.976$

Example: $X \sim N(15, 4)$



The probability that X is between 11 and 19 is approximately $=$ 0.68

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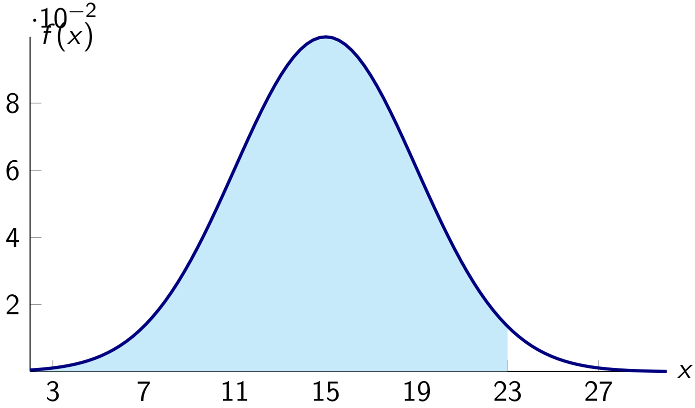

The probability that X is is less than 23 is approximately $= 0.975$

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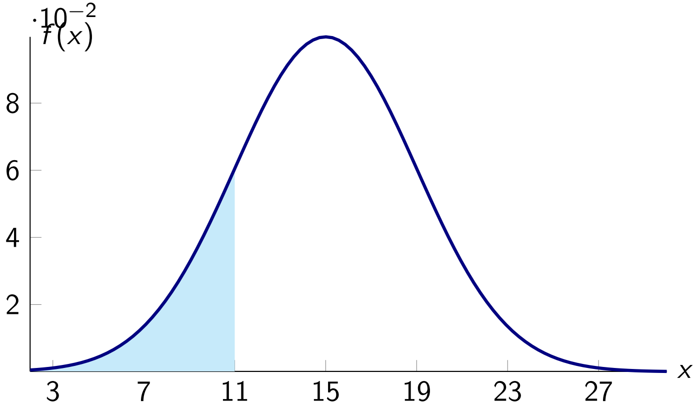
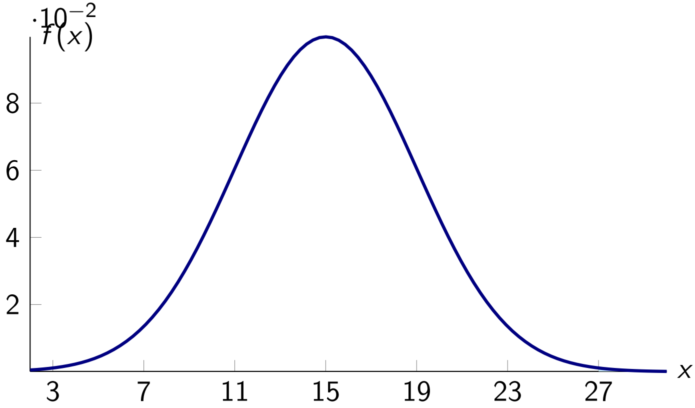


The probability that X is less than 11 is approximately $= 0.16$

Example: $X \sim N(15, 4)$



The probability that X is greater than 27 is approximately $= 0$

Example: Finding μ given probability and σ

Variable X is a normal random variable with standard deviation 3. If the probability that X is between 7 and 19 is 0.95, then the expected value of X is approximately

Example: Finding μ given probability and σ

Variable X is a normal random variable with standard deviation 3. If the probability that X is between 7 and 19 is 0.95, then the expected value of X is approximately 13

Example: Finding μ given probability and σ

Variable X is a normal random variable with standard deviation 3. If the probability that X is greater than 16 is 0.975, then the expected value of X is approximately

Example: Finding μ given probability and σ

Variable X is a normal random variable with standard deviation 3. If the probability that X is greater than 16 is 0.975, then the expected value of X is approximately 22

Example: Finding σ given probability and μ

Variable X is a normal random variable with expected value 100. If the probability that X is greater than 130 is 0.025, then the standard deviation of X is approximately

Example: Finding σ given probability and μ

Variable X is a normal random variable with expected value 100. If the probability that X is greater than 130 is 0.025, then the standard deviation of X is approximately 15

Comparing two distributions

If X is normal with expected value 100 and standard deviation 2, and Y is normal with expected value 105 and standard deviation 10, is X or is Y more likely to exceed 105?

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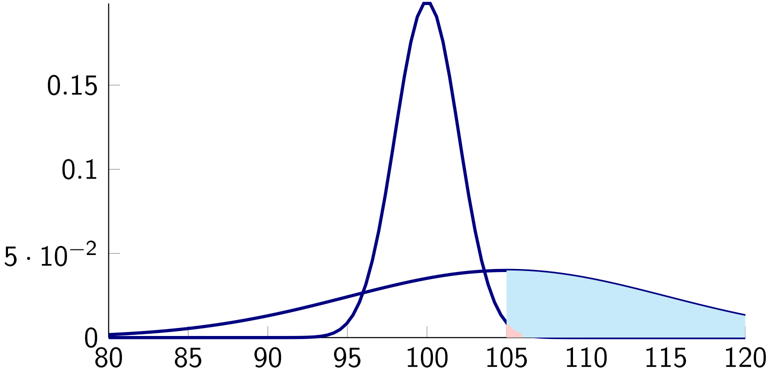
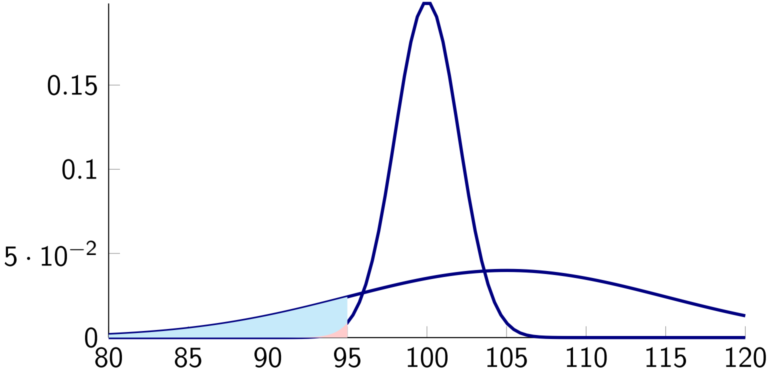
1. $P(X > 105) \approx 0.0062, P(Y > 105) = 0.5$



Comparing two distributions

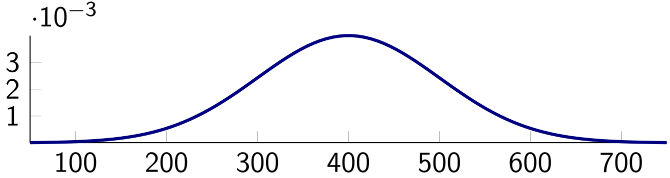
If X is normal with expected value 100 and standard deviation 2, and Y is normal with expected value 105 and standard deviation 10, is X or is Y more likely to be less than 95

1. $P(X < 95) \approx 0.0062, P(Y < 95) \approx 0.16$



Example

The scores on a particular job aptitude test are normal with expected value 400 and standard deviation 100. If a company will consider only those applicants scoring in the top 5 percent, determine whether it should consider one whose score is



1. 400 NO
2. 450 NO
3. 500 NO
4. 600 YES

Section summary

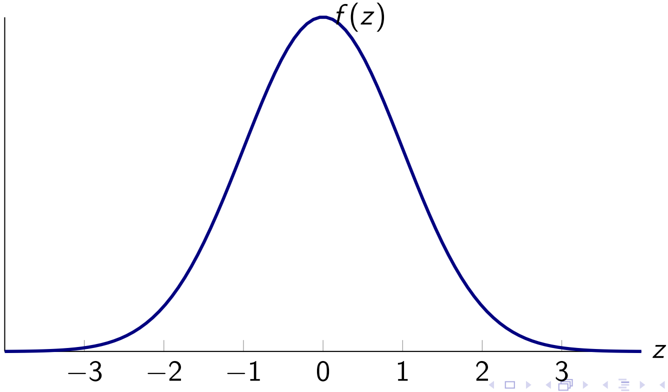
- Approximation rule and its applications.

Standard Normal Distribution: $\mu = 0$ and $\sigma = 1$

- A normal random variable having mean value 0 and standard deviation 1 is called a standard normal random variable, and its density curve is called the standard normal curve.

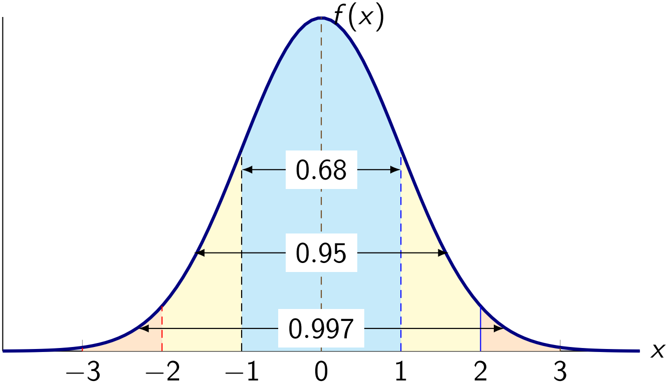
Standard Normal Distribution: $\mu = 0$ and $\sigma = 1$

- A normal random variable having mean value 0 and standard deviation 1 is called a standard normal random variable, and its density curve is called the standard normal curve.
- We represent a standard normal random variable by Z.

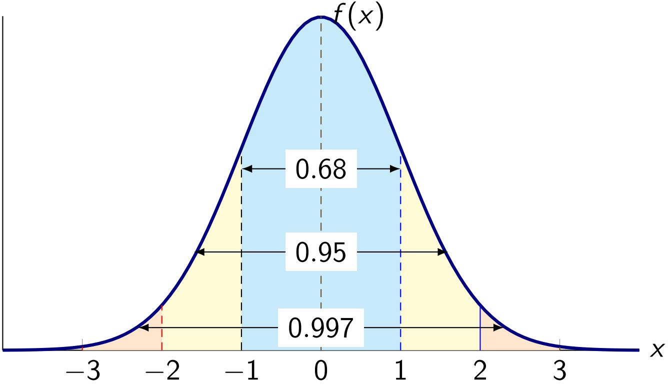


Approximation rule for standard normal

- Between −1 and +1 with approx. probability 0.68
- Between −2 and +2 with approx. probability 0.95
- Between −3 and +3 with approx. probability 0.997

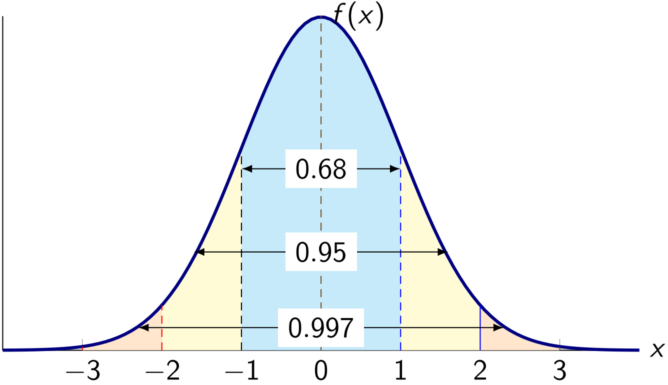


Application of approximate rule to standard normal probabilities



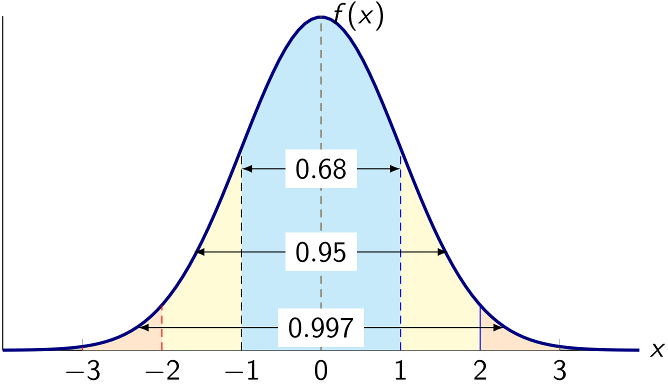
1. $P(-2 < Z < 2) =$

Application of approximate rule to standard normal probabilities



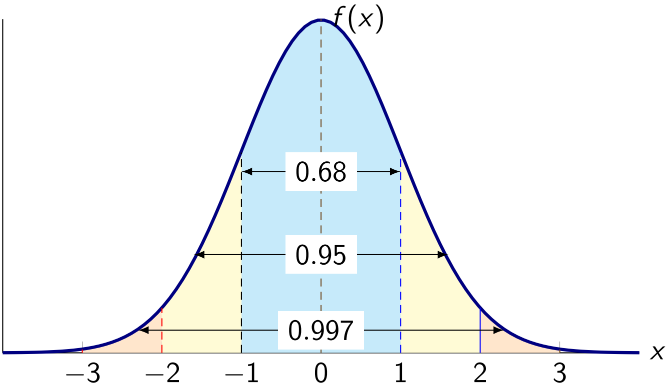
1. $P(-2 < Z < 2) = 0.95$
2. $P(Z > -1) =$

Application of approximate rule to standard normal probabilities



1. $P(-2 < Z < 2) = 0.95$
2. $P(Z > -1) = 0.84$
3. $P(Z > 1) =$

Application of approximate rule to standard normal probabilities



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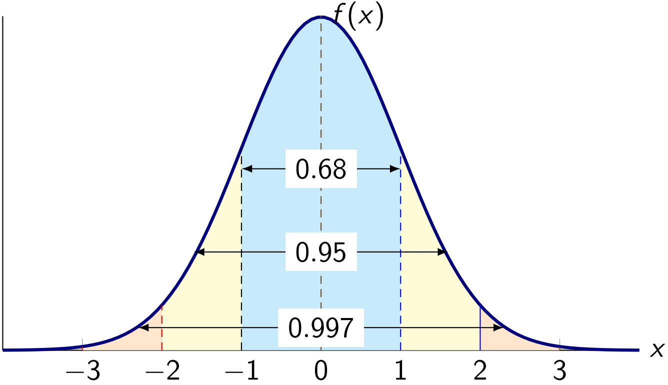

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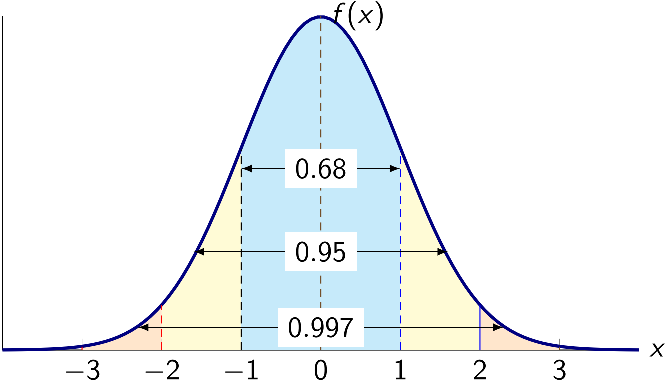
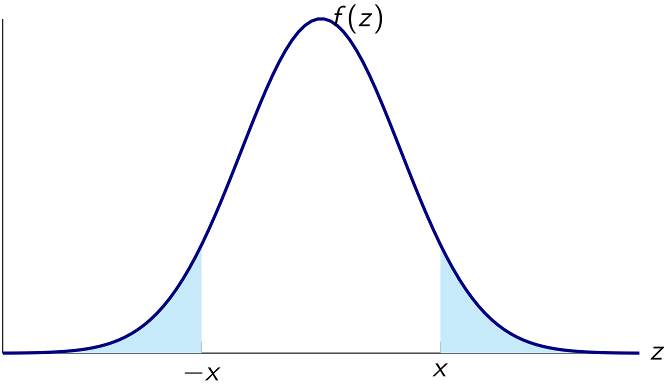
5. $P(Z \leq 2) =$

Application of approximate rule to standard normal probabilities



1. $P(-2 < Z < 2) = 0.95$
2. $P(Z > -1) = 0.84$
3. $P(Z > 1) = 0.16$
4. $P(Z > 3) = 0$
5. $P(Z \leq 2) = 0.975$

Property 1: $P(Z < -x) = 1 - P(Z < x)$



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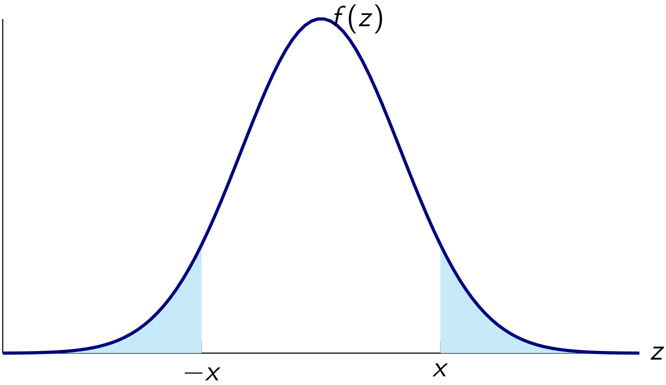

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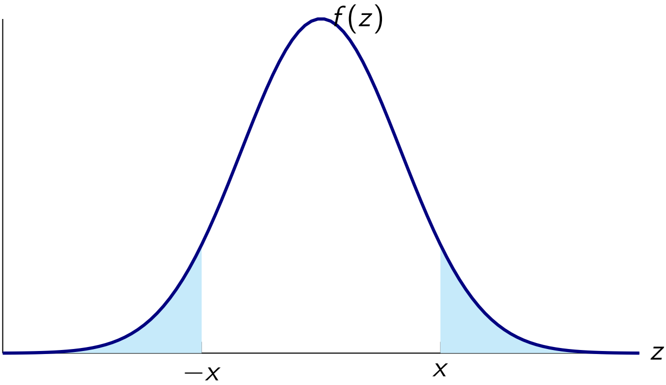
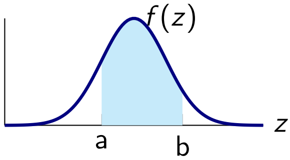


- $P(Z < -x) = P(Z > x) = 1 - P(Z < x)$

Property 2: $P(a < Z < b) = P(Z < b) - P(Z < a)$



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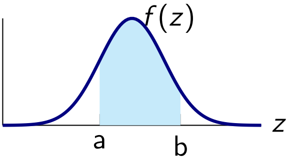
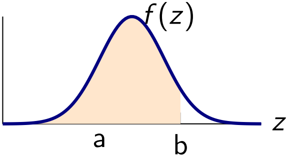
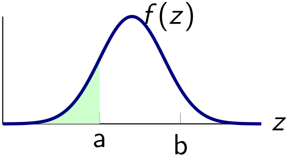


This can be expressed as difference of the areas:





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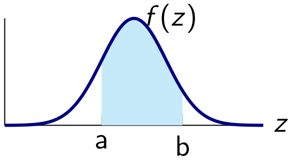

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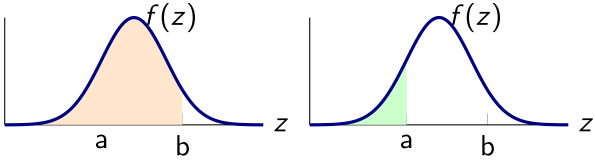
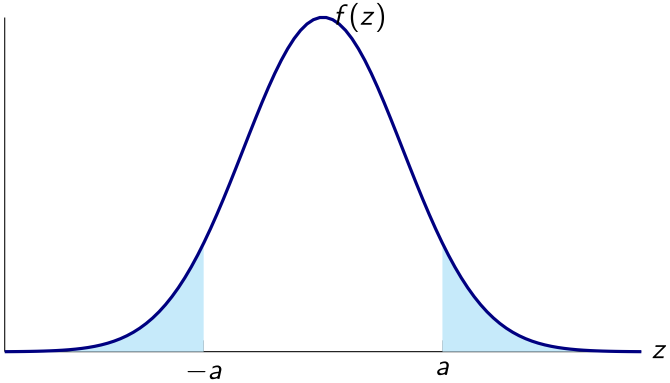
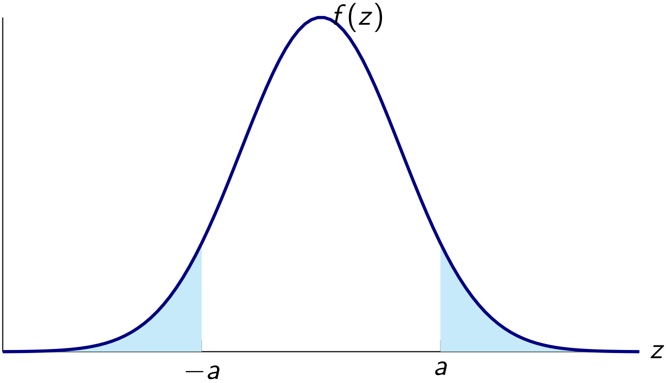


- $P(a < Z < b) = P(Z < b) - P(Z < a)$

Property 3: $P(|Z| > a) = 2(1 - P(Z < a))$



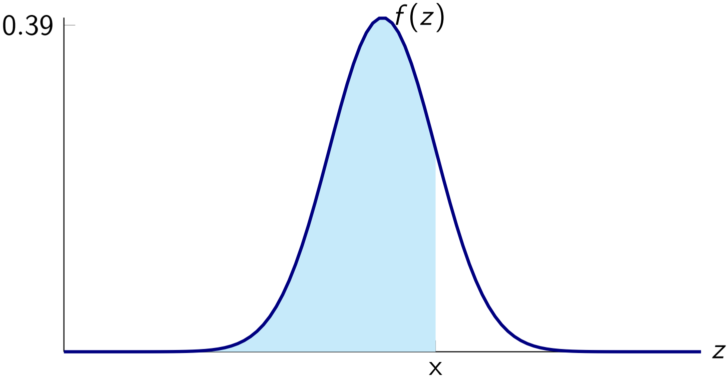
Property 3: $P(|Z| > a) = 2(1 - P(Z < a))$



- $P(|Z| > a) = P(Z > a) + P(Z < -a) = 2P(Z > a) = 2(1 - P(Z < a))$

Standard Normal table

- For each nonnegative value of x, the table specifies the probability that Z is less than x, i.e. $P(Z \leq x)$.



Portion of Standard Normal table

z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**
0	$\boxed{0.5}$	0.504	0.508	0.512	0.516	0.5199	0.5239
0.1	0.5398	0.5438	$\boxed{0.5478}$	0.5517	0.5557	0.5596	0.5636
0.2	0.5793	0.5832	0.5871	0.591	0.5948	0.5987	0.6026
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406
0.4	0.6554	0.6591	0.6628	0.6664	0.67	0.6736	0.6772
0.5	0.6915	0.695	0.6985	0.7019	0.7054	0.7088	0.7123
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454
0.7	0.758	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764
0.8	0.7881	0.791	0.7939	0.7967	$\boxed{0.7995}$	0.8023	0.8051
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315
1	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.877
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962

Standard Normal table

z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
0	0.5	0.504	0.508	0.512	0.516	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.591	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.648	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.67	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.695	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.719	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.758	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.791	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.834	0.8365	0.8389
1	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.877	0.879	0.881	0.883
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.898	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.937	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.975	0.9756	0.9761	0.9767
2	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.983	0.9834	0.9838	0.9842	0.9846	0.985	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.989
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.992	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.994	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.996	0.9961	0.9962	0.9963	0.9964

Using Normal tables to compute probabilities

Using Normal tables to compute probabilities

1. $P(Z < 2.2) =$

Using Normal tables to compute probabilities

1. $P(Z < 2.2) = 0.9861$

z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
2.2	$\boxed{0.9861}$	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.989

Using Normal tables to compute probabilities

1. $P(Z < 2.2) = 0.9861$

z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
2.2	$\boxed{0.9861}$	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.989

2. $P(Z > 1.1) =$

Using Normal tables to compute probabilities

1. $P(Z < 2.2) = 0.9861$

z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
2.2	$\boxed{0.9861}$	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.989

2. $P(Z > 1.1) = 0.1357$

z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
1.1	$\boxed{0.8643}$	0.8665	0.8686	0.8708	0.8729	0.8749	0.877	0.879	0.881	0.883

Using Normal tables to compute probabilities

1. $P(Z < 2.2) = 0.9861$

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
2.2	$\boxed{0.9861}$	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.989

2. $P(Z > 1.1) = 0.1357$

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.1	$\boxed{0.8643}$	0.8665	0.8686	0.8708	0.8729	0.8749	0.877	0.879	0.881	0.883

3. $P(0 < Z < 2) =$

Using Normal tables to compute probabilities

1. $P(Z < 2.2) = 0.9861$

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
2.2	$\boxed{0.9861}$	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.989

2. $P(Z > 1.1) = 0.1357$

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.1	$\boxed{0.8643}$	0.8665	0.8686	0.8708	0.8729	0.8749	0.877	0.879	0.881	0.883

3. $P(0 < Z < 2) = 0.4772$

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.2	$\boxed{0.8849}$	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.898	0.8997	0.901

Using Normal tables to compute probabilities

Using Normal tables to compute probabilities

4. $P(-0.9 < Z < 1.2) =$

Using Normal tables to compute probabilities

4. $P(-0.9 < Z < 1.2) = 0.7008$

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.834	0.8365	0.8389
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.898	0.8997	0.9015

$$P(-0.9 < Z < 1.2) = P(Z < 1.2) - P(Z < -0.9) =$$

Using Normal tables to compute probabilities

4. $P(-0.9 < Z < 1.2) = 0.7008$

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.834	0.8365	0.8389
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.898	0.8997	0.9015

$P(-0.9 < Z < 1.2) = P(Z < 1.2) - P(Z < -0.9) =$
$P(Z < 1.2) - P(Z > 0.9) =$

Using Normal tables to compute probabilities

4. $P(-0.9 < Z < 1.2) = 0.7008$

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.9	$\boxed{0.8159}$	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.834	0.8365	0.8389
1.2	$\boxed{0.8849}$	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.898	0.8997	0.9015

$$P(-0.9 < Z < 1.2) = P(Z < 1.2) - P(Z < -0.9) =$$
$$P(Z < 1.2) - P(Z > 0.9) =$$
$$P(Z < 1.2) + P(Z < 0.9) - 1 =$$

Using Normal tables to compute probabilities

4. $P(-0.9 < Z < 1.2) = 0.7008$

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.834	0.8365	0.8389
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.898	0.8997	0.9015

$$P(-0.9 < Z < 1.2) = P(Z < 1.2) - P(Z < -0.9) = P(Z < 1.2) - P(Z > 0.9) = P(Z < 1.2) + P(Z < 0.9) - 1 = 0.8849 + 0.8159 - 1 = 0.7008$$

Using Normal tables to compute probabilities

Using Normal tables to compute probabilities

5. $P(Z > -1.96) =$

Using Normal tables to compute probabilities

5. $P(Z > -1.96) = 0.9750$

z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	$\boxed{0.975}$	0.9756	0.9761	0.9767

Using Normal tables to compute probabilities

5. $P(Z > -1.96) = 0.9750$

z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	$\boxed{0.975}$	0.9756	0.9761	0.9767

6. $P(Z < -0.72) =$

Using Normal tables to compute probabilities

5. $P(Z > -1.96) = 0.9750$

z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	$\boxed{0.975}$	0.9756	0.9761	0.9767

6. $P(Z < -0.72) = 0.2358$

z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
0.7	0.758	0.7611	$\boxed{0.7642}$	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852

Using Normal tables to compute probabilities

Using Normal tables to compute probabilities

7. $P(|Z| < 1.64) =$

Using Normal tables to compute probabilities

7. $P(|Z| < 1.64) = 0.899$

Using Normal tables to compute probabilities

7. $P(|Z| < 1.64) = 0.899$
8. $P(|Z| > 1.20) =$

Using Normal tables to compute probabilities

7. $P(|Z| < 1.64) = 0.899$
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Using Normal tables to compute probabilities

7. $P(|Z| < 1.64) = 0.899$
8. $P(|Z| > 1.20) = 0.2302$
9. $P(-2.2 < Z < 1.2) =$

Using Normal tables to compute probabilities

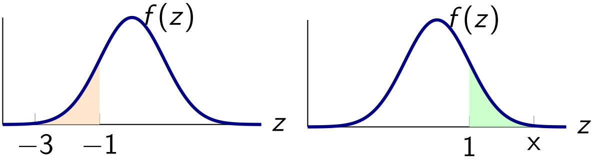
7. $P(|Z| < 1.64) = 0.899$
8. $P(|Z| > 1.20) = 0.2302$
9. $P(-2.2 < Z < 1.2) = 0.8710$

z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
1.2	$\boxed{0.8849}$	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.898	0.8997	0.9015
1.6	0.9452	0.9463	0.9474	0.9484	$\boxed{0.9495}$	0.9505	0.9515	0.9525	0.9535	0.9545
2.2	$\boxed{0.9861}$	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.989

Using Normal tables to compute unknown x

Find the value of x:

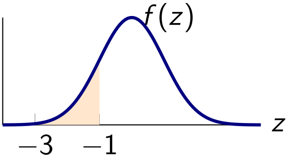
$$P(-3 < Z < -1) = P(1 < Z < x)$$

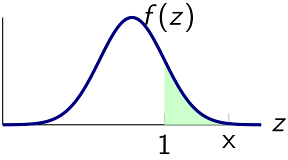


Using Normal tables to compute unknown x

Find the value of x:

$$P(-3 < Z < -1) = P(1 < Z < x)$$





solution: $x = 3$

Examples

Find value of x

1. $P(Z > x) = 0.05$, $x = 1.65$
2. $P(Z > x) = 0.005$, $x = 2.58$
3. $P(|Z| < x) = 0.75$, $2P(Z < x) = 1.75; x = 1.15$

z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	$\boxed{0.8749}$	0.877	0.879	0.881	0.883
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	$\boxed{0.9505}$	0.9515	0.9525	0.9535	0.9545
2.5	0.9938	0.994	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	$\boxed{0.9951}$	0.9952

Section summary

- Standard normal distribution
- Normal tables
- Using normal tables to compute probabilities.

Standardizing to Find Normal Probabilities

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- Start by converting x into a z-score

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$$Z = \frac{X - \mu}{\sigma}$$

The value of the standardized variable tells us how many standard deviations the original variable is from its mean.

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- We can compute any probability statement about X by writing an equivalent statement in terms of Z

Standardizing to Find Normal Probabilities

- Start by converting x into a z-score

$$Z = \frac{X - \mu}{\sigma}$$

The value of the standardized variable tells us how many standard deviations the original variable is from its mean.

- We can compute any probability statement about X by writing an equivalent statement in terms of Z

$$P(X < a) = P\left(\frac{X - \mu}{\sigma} < \frac{a - \mu}{\sigma}\right) = P\left(Z < \frac{a - \mu}{\sigma}\right)$$

Example: Distribution of IQ scores

Suppose IQ examination scores for students of a particular school are normally distributed with mean value 140 and standard deviation 10.

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1. What is the probability a randomly chosen student has a score greater than 130?

Example: Distribution of IQ scores

Suppose IQ examination scores for students of a particular school are normally distributed with mean value 140 and standard deviation 10.

1. What is the probability a randomly chosen student has a score greater than 130?

$$P(X > 130) = P(Z > -1) = 0.84$$

z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**
1	$\boxed{0.8413}$	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554

Example: Distribution of IQ scores

Suppose IQ examination scores for students of a particular school are normally distributed with mean value 140 and standard deviation 10.

1. What is the probability a randomly chosen student has a score greater than 130?

$$P(X > 130) = P(Z > -1) = 0.84$$

z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**
1	$\boxed{0.8413}$	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554

2. What is the probability a randomly chosen student has a score between 120 and 150?

$$P(110 < X < 150) = P(-2 < Z < 1) = 0.8185$$

z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
2	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817

$|X - \mu| > a\sigma$

Let X be normal with mean μ and standard deviation σ. Find

$|X - \mu| > a\sigma$

Let X be normal with mean μ and standard deviation σ. Find

1. $P(|X - \mu| > \sigma) = P(|Z| > 1) = 0.3174$

$|X - \mu| > a\sigma$

Let X be normal with mean μ and standard deviation σ. Find

1. $P(|X - \mu| > \sigma) = P(|Z| > 1) = 0.3174$
2. $P(|X - \mu| > 2\sigma) = P(|Z| > 2) = 0.0456$

$|X - \mu| > a\sigma$

Let X be normal with mean μ and standard deviation σ. Find

1. $P(|X - \mu| > \sigma) = P(|Z| > 1) = 0.3174$
2. $P(|X - \mu| > 2\sigma) = P(|Z| > 2) = 0.0456$
3. $P(|X - \mu| > 3\sigma) = P(|Z| > 3) = 0.0026$

$|X - \mu| > a\sigma$

Let X be normal with mean μ and standard deviation σ. Find

1. $P(|X - \mu| > \sigma) = P(|Z| > 1) = 0.3174$
2. $P(|X - \mu| > 2\sigma) = P(|Z| > 2) = 0.0456$
3. $P(|X - \mu| > 3\sigma) = P(|Z| > 3) = 0.0026$

- The statement $|X - \mu| > a\sigma$ is, in terms of the standardized variable Z, is equivalent to the statement $|Z| > a$

Example: Transforming normal to Standard normal

X is normal with mean 10 and standard deviation 3, find

Example: Transforming normal to Standard normal

X is normal with mean 10 and standard deviation 3, find

1. $P(X > 12) =$

Example: Transforming normal to Standard normal

X is normal with mean 10 and standard deviation 3, find

1. $P(X > 12) = P(Z > 0.67) = 0.2514$

Example: Transforming normal to Standard normal

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3. $P(8 < X < 11) =$

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4. $P(X > 7) =$

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Example: Exam scores

The scores on an entrance exam were normally distributed with mean 500 and standard deviation 90.

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z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
2.2	0.9861	0.9864	$\boxed{0.9868}$	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.989

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$P(X > 700) = P(Z > 2.22) = 0.0132.$

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$P(X > 700) = P(Z > 2.22) = 0.0132$. Thus, the percentage who scored higher than 700 is 1.32%

Example: lifetime of automobile tires

The life of a certain automobile tire is normally distributed with mean 35,000 miles and standard deviation 5000 miles.

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1. What proportion of such tires last between 30,000 and 40,000 miles?

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1. What proportion of such tires last between 30,000 and 40,000 miles? $P(30 < X < 40) =$

Example: lifetime of automobile tires

The life of a certain automobile tire is normally distributed with mean 35,000 miles and standard deviation 5000 miles. Let X be the life of the tire in thousands of miles.

1. What proportion of such tires last between 30,000 and 40,000 miles? $P(30 < X < 40) = P(-1 < Z < 1) = 0.6826$

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1. What proportion of such tires last between 30,000 and 40,000 miles? $P(30 < X < 40) = P(-1 < Z < 1) = 0.6826$
2. What proportion of such tires last over 40,000 miles?

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3. What proportion last over 50,000 miles? $P(X > 50) =$

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2. What proportion of such tires last over 40,000 miles? $P(X > 40) = P(Z > 1) = 0.1587$
3. What proportion last over 50,000 miles? $P(X > 50) = P(Z > 3) = 0.0013$

Example: Pulse rate

The pulse rate of young adults is normally distributed with a mean of 72 beats per minute and a standard deviation of 9.5 beats per minute. If the requirements for the military rule out anyone whose rate is over 95 beats per minute, what percentage of the population of young adults does not meet this standard?

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- Let X be pulse rate of young adults
- $P(X > 95); X \sim N(72, 9.5)$

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- Let X be pulse rate of young adults
- $P(X > 95); X \sim N(72, 9.5)$
- $P(X > 95) =$

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- $P(X > 95); X \sim N(72, 9.5)$
- $P(X > 95) = P(Z > 2.42) = 0.0078$

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- Let X be pulse rate of young adults
- $P(X > 95); X \sim N(72, 9.5)$
- $P(X > 95) = P(Z > 2.42) = 0.0078$
- Thus 0.78% of the population of young adults do not meet the military standard.

Example: Manufacturing bolts

The bolts produced by a manufacturer are specified to be between 1.09 and 1.11 inches in diameter. If the production process results in the diameter of bolts being a normal random variable with mean 1.10 inches and standard deviation 0.005 inch, what percentage of bolts do not meet the specifications?

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- Let X be diameter of the bolts in inches.

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- Let X be diameter of the bolts in inches.
- $X \sim N(1.1, 0.005)$

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- Let X be diameter of the bolts in inches.
- $X \sim N(1.1, 0.005)$
- $P(1.09 < X < 1.11) =$

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- Let X be diameter of the bolts in inches.
- $X \sim N(1.1, 0.005)$
- $P(1.09 < X < 1.11) = P(-2 < Z < 2) = 0.9544$

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- Let X be diameter of the bolts in inches.
- $X \sim N(1.1, 0.005)$
- $P(1.09 < X < 1.11) = P(-2 < Z < 2) = 0.9544$
- Thus the proportion of bolts that do not meet the specifications is $(1 - 0.9544) \times 100\% = 4.56\%$

Example: Junking car battery

You are planning on junking your old car after it runs an additional 20,000 miles. The battery on this car has just failed, and you must decide which of two types of batteries, costing the same amount, to purchase. After some research you have discovered that the lifetime of the first battery is normally distributed with mean life 24,000 and standard deviation 6000 miles, and the lifetime of the second battery is normally distributed with mean life 22,000 and standard deviation 2000 miles.

Example: Junking car battery-solution

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- If all you care about is that the battery purchased lasts at least 20,000 miles, which one should you buy?

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- If all you care about is that the battery purchased lasts at least 20,000 miles, which one should you buy?
- Let X =lifetime of first battery and let Y = lifetime of second battery.
- $P(X > 20) = 0.7486$;

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- If all you care about is that the battery purchased lasts at least 20,000 miles, which one should you buy?
- Let X =lifetime of first battery and let Y = lifetime of second battery.
- $P(X > 20) = 0.7486; P(Y > 20) = 0.8413$

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- Let X =lifetime of first battery and let Y = lifetime of second battery.
- $P(X > 20) = 0.7486; P(Y > 20) = 0.8413$
- Since second battery has higher probability of lasting beyond 20,000 miles, purchase the second battery

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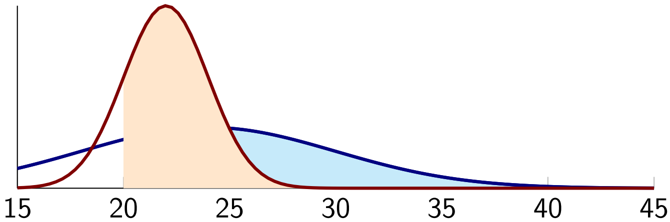
- If all you care about is that the battery purchased lasts at least 20,000 miles, which one should you buy?
- Let X =lifetime of first battery and let Y = lifetime of second battery.
- $P(X > 20) = 0.7486; P(Y > 20) = 0.8413$
- Since second battery has higher probability of lasting beyond 20,000 miles, purchase the second battery
- Would your answer change if you wanted the battery to last 21,000 miles?


15
20
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40
45

Section summary

- Standardizing a normal random variable
- Applications.

Additive Property

Result

The sum of independent normal random variables is also a normal random variable. Suppose X and Y are independent normal random variables with respective parameters μ_X, σ_X and μ_Y, σ_Y, then $X + Y$ also will be normal with mean $\mu_X + \mu_Y$ and standard deviation $\sqrt{\sigma_X^2 + \sigma_Y^2}$.

Application of additive property: lifetime of two bulbs

Suppose the amount of time a light bulb works before burning out is a normal random variable with mean 400 hours and standard deviation 40 hours. If an individual purchases two such bulbs, one of which will be used as a spare to replace the other when it burns out, what is the probability that the total life of the bulbs will exceed 750 hours?

Application of additive property: lifetime of two bulbs

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- $P(X + Y) > 750 = ?$

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- $P(X+Y) > 750 = ?$
- $X+Y \sim N(800, \sqrt{3200})$

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- $P(X+Y) > 750 = ?$
- $X + Y \sim N(800, \sqrt{3200})$
- $P(X+Y > 750) = P(Z < 0.884) = 0.81$

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- $P(X + Y) > 750 =?$
- $X + Y \sim N(800, \sqrt{3200})$
- $P(X + Y > 750) = P(Z < 0.884) = 0.81$
- Therefore, there is an 81 percent chance that the total life of the two bulbs exceeds 750 hours.

Application of additive property: Computing rainfall over two years

The annual rainfall in Guwahati, is normally distributed with mean 67 inches and standard deviation 8 inches.
Let X be annual rainfall in inches in Guwahati.

Application of additive property: Computing rainfall over two years

The annual rainfall in Guwahati, is normally distributed with mean 67 inches and standard deviation 8 inches.
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1. What is the probability that this year's rainfall exceeds 70 inches?

Application of additive property: Computing rainfall over two years

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Let X be annual rainfall in inches in Guwahati.

1. What is the probability that this year's rainfall exceeds 70 inches?
   $P(X > 70) = P(Z > 0.375) = 0.3538$
2. What is the probability that the sum of the next 2 years? rainfall exceeds 140 inches?

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   let Y be sum of next two years rainfall. $Y \sim N(134, 11.313)$.

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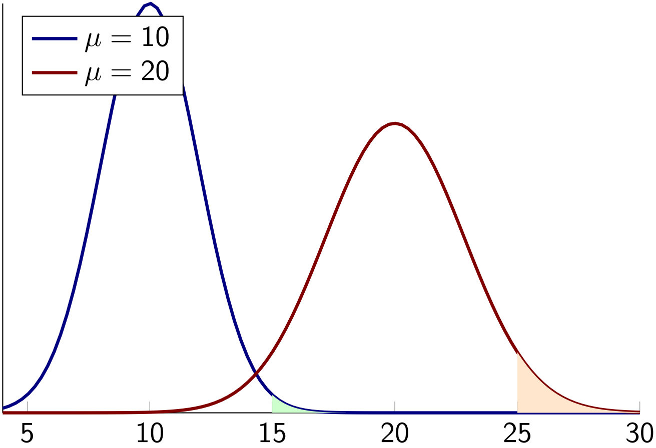
2. What is the probability that the sum of the next 2 years? rainfall exceeds 140 inches?
   let Y be sum of next two years rainfall. $Y \sim N(134, 11.313)$.
   $P(Y > 140) = P(Z > 0.53) = 0.297$

Comparing distributions

Let X_1 and X_2 be independent normal random variables, each having mean 10 and variance 4. Which probability is larger

1. $P(X_1 > 15)$
2. $P(X_1 + X_2 > 25)$
3. $P(X_1 + X_2 > 30)$


$\mu = 10$
$\mu = 20$
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Section summary

1. Additive property of normal random variables and its applications.

Percentiles

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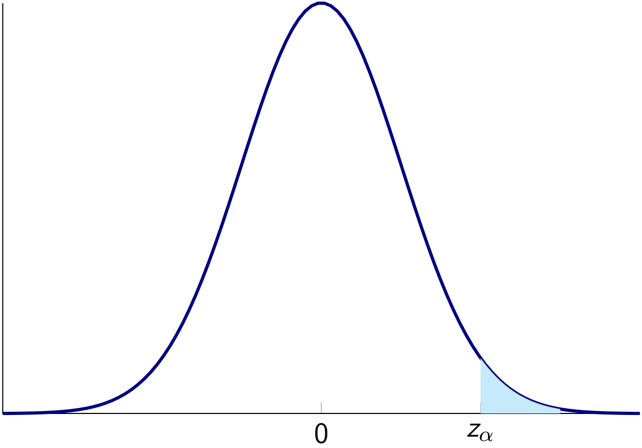
$$P(Z > z_\alpha) = \alpha$$

Percentiles

- The previous problems ask us to find the probability associated with being larger or smaller than some value.
- Some problems require working in the opposite direction that starts with the probability
- For any α between 0 and 1, we define z_α to be that value for which
$$P(Z > z_\alpha) = \alpha$$
- z_α is the $100(1-\alpha)$ percentile of the standard normal distribution.

Percentile



$$P(Z > z_\alpha) = \alpha$$

$z_{0.025}$

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- What is $z_{0.025}$? $P(Z < z_{0.025}) = 1 - P(Z > z_{0.025}) = 0.975.$

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z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
0	0.5	0.504	0.508	0.512	0.516	0.5199	0.5239	0.5279	0.5319	0.5359
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	$\boxed{0.975}$	0.9756	0.9761	0.9767

- Search for entry 0.975 in table and find value of x corresponding to this value. We see that $z_{0.025} = 1.96$

$z_{0.025}$

- What is $z_{0.025}$? $P(Z < z_{0.025}) = 1 - P(Z > z_{0.025}) = 0.975$.

z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
0	0.5	0.504	0.508	0.512	0.516	0.5199	0.5239	0.5279	0.5319	0.5359
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	$\boxed{0.975}$	0.9756	0.9761	0.9767

- Search for entry 0.975 in table and find value of x corresponding to this value. We see that $z_{0.025} = 1.96$
- That is, 2.5 percent of the time a standard normal variable will exceed 1.96.

$z_{0.025}$

- What is $z_{0.025}$? $P(Z < z_{0.025}) = 1 - P(Z > z_{0.025}) = 0.975$.

z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
0	0.5	0.504	0.508	0.512	0.516	0.5199	0.5239	0.5279	0.5319	0.5359
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	$\boxed{0.975}$	0.9756	0.9761	0.9767

- Search for entry 0.975 in table and find value of x corresponding to this value. We see that $z_{0.025} = 1.96$
- That is, 2.5 percent of the time a standard normal variable will exceed 1.96.
- Since 97.5 percent of the time a standard normal random variable will be less than 1.96, we say that 1.96 is the 97.5 percentile of the standard normal distribution.

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- In general, since $100(1 - \alpha)$ percent of the time a standard normal random variable will be less than z_α, we call z_α the $100(1 - \alpha)$ percentile of the standard normal distribution.

$z_{0.05}$

- What is $z_{0.05}$?
- For the value 0.95, we do not find this exact value. Rather, we see that $P(Z < 1.64) = 0.9495$ and $P(Z < 1.65) = 0.9505$
- Therefore, it would seem that $z_{0.05}$ lies roughly halfway between 1.64 and 1.65, and so we could approximate it by 1.645.

z	**0**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.0**
1.6	0.9452	0.9463	0.9474	0.9484	$\boxed{0.9495}$	$\boxed{0.9505}$	0.9515	0.9525	0.9535	0.9

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1.6	0.9452	0.9463	0.9474	0.9484	$\boxed{0.9495}$	$\boxed{0.9505}$	0.9515	0.9525	0.9535	0.9

- The values of $z_{0.10}, z_{0.05}, z_{0.025}, z_{0.01}$, and $z_{0.005}$, are of particular importance in statistics. Their values are as follows:

$$z_{0.10} = 1.282, z_{0.025} = 1.96, z_{0.005} = 2.576$$

$$z_{0.05} = 1.645, z_{0.01} = 2.326$$

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- Hence, the desired value of $a = 5(1.645) + 40 = 48.225$

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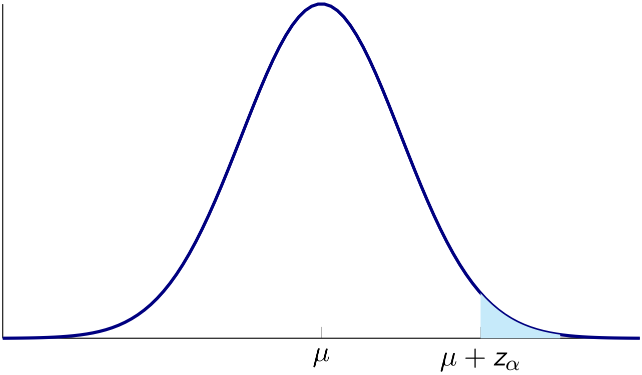
- $\frac{a-100}{14.2} = z_{0.01} = 2.33$
- $x = 14(2.33) + 100 = 133.086$
- That is, the top 1 percent consists of all those having scores above 134.

In general

Let X be a normal random variable with mean μ and standard deviation σ

$$P(X > \mu + \sigma z_\alpha) = \alpha$$



Summary

1. Approximation rule
2. Standard Normal variable
3. Standardizing and applications.
4. Adding two normal variables.
5. Percentiles of normal distribution